

Based on the Pauli exclusion principle, which electron subshell configurations cannot exist?

- I. $3s$ $\boxed{\uparrow \uparrow}$
 II. $2p^7$
 III. $6d$ $\boxed{\uparrow \downarrow} \boxed{\uparrow \downarrow} \boxed{\uparrow} \boxed{\uparrow} \boxed{\uparrow}$
 IV. $4f^{14}$

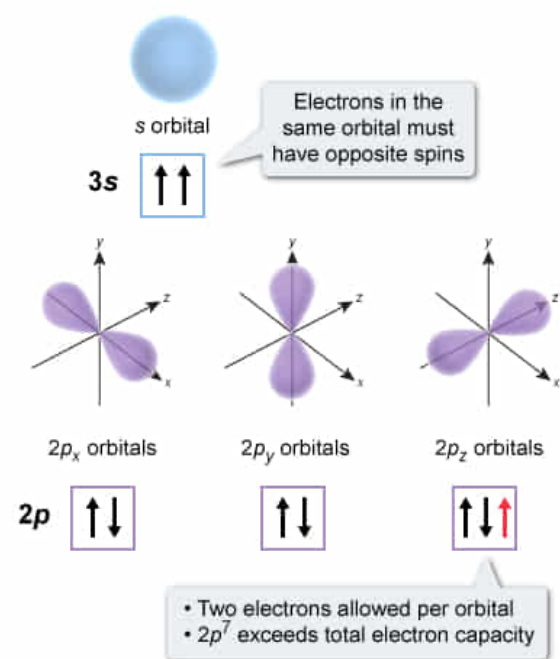
- ✓ ☒ A. I and II only [82%]
☐ B. I, II, and III only [8%]
☐ C. II and III only [4%]
☐ D. II, III, and IV only [3%]

Correct

82% answered correctly

Explanation:

Invalid electron subshell configurations



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An **electron configuration** sequentially lists the placement of all electrons within the **shells and subshells** of an atom or ion in **order** of increasing energy, according to the Aufbau principle. Subshells are labeled by type (**s**, **p**, **d**, **f**), which indicates the kind of **orbitals** present within the subshell. Accordingly, an s subshell contains one s orbital, a p subshell contains three p orbitals, a d subshell contains five d orbitals, and an f subshell contains seven f orbitals.

The **Pauli exclusion principle** states that each orbital within a subshell can hold a

maximum of two electrons, but two electrons in the same orbital must have **opposite spins**. Therefore, *s*, *p*, *d*, and *f* subshells can hold a maximum of 2, 6, 10, and 14 electrons, respectively. In schematic diagrams, orbitals are often represented by blanks or boxes, and electrons are represented by arrows. The spin of each electron is indicated by the orientation of the arrow (up or down).

Accordingly, a subshell represented as $3s \begin{array}{|c|c|} \hline \uparrow & \uparrow \\ \hline \end{array}$ (**Number I**) cannot exist because the electrons in the *s* orbital have the same spin. Likewise, a subshell indicated by $2p^7$ (**Number II**) also cannot exist because a *p* subshell holds a maximum of 6 electrons (two electrons in each of the three *p* orbitals).

(**Number III**) $6d \begin{array}{|c|c|} \hline \uparrow & \downarrow \\ \hline \end{array} \begin{array}{|c|c|} \hline \uparrow & \downarrow \\ \hline \end{array} \begin{array}{|c|} \hline \uparrow \\ \hline \end{array} \begin{array}{|c|} \hline \uparrow \\ \hline \end{array} \begin{array}{|c|} \hline \uparrow \\ \hline \end{array}$ is a valid subshell configuration. A *d* subshell contains five *d* orbitals of equal energy that together can accommodate up to 10 electrons. Because the *d* orbitals are equivalent in energy, paired electrons of opposite spin can be placed in any of the five orbitals without distinction.

(**Number IV**) $4f^{14}$ is a correct subshell configuration. An *f* subshell contains seven *f* orbitals of equal energy that together can accommodate up to 14 electrons in total.

Educational objective:

In an electron configuration, subshells are labeled by type as *s*, *p*, *d*, or *f*, which contain one *s* orbital, three *p* orbitals, five *d* orbitals, and seven *f* orbitals, respectively. According to the Pauli exclusion principle, each orbital within a subshell can hold a maximum of two electrons, but two electrons in the same orbital must have opposite spins.

Time spent:0 sec

Subject:

General Chemistry

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System:

Atomic Theory and Chemical Composition